Chapter 12, Problem 48P

Problem

A balanced, positive-sequence wye-connected source has $\mathbf{V}_{an} = 240\angle 0^\circ$ V rms and supplies an unbalanced delta-connected load via a transmission line with impedance 2 + *j*3 Ω per phase.

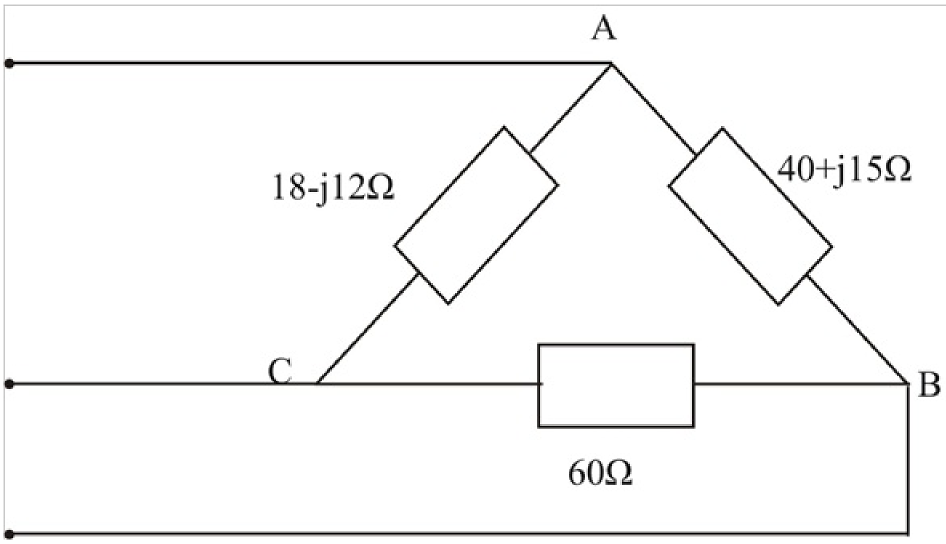
(a) Calculate the line currents if **Z***AB* = 40 + *j*15 Ω, **Z***BC* = 60 Ω, **Z***CA* = 18 − *j*12 Ω.

(b) Find the complex power supplied by the source.

Step-by-step solution

Step 1 of 4

(A) We first convert the delta load to its equivalent WYE – load as shown below.



Step 2 of 4

$$Z_A = \frac{(40+j15)(18-j12)}{118+j3} = 7.577 - j1.923$$

$$Z_B = \frac{60(40+j15)}{118+j3} = 20.52 + j7.105$$

$$Z_C = \frac{60(18-j12)}{118+j3} = 8.992 - j6.3303$$

The system becomes that shown below.

We apply KVL to the loops.

For Mesh 1,

$$-240 + 240\angle -120^\circ + I_1(2Z_1 + Z_A + Z_B) - I_2(Z_B + Z_1) = 0$$

Or, $(32.097 + j11.13)I_1 - (22.52 + j10.105)I_2 = 360 + j20.785 \rightarrow (1)$

Step 3 of 4

For Mesh 2,

$$240\angle 120° - 240\angle -120° - I_1(Z_B+Z_C) + I_2(2Z_1+Z_B+Z_C) = 0$$

$$-(22.52+j10.105)I_1 + (33.51+56.775)I_2 = -j415.69 \rightarrow (2)$$

Solving (1) & (2) gives,

$$I_1 = 23.75 - j5.328$$

$$I_2 = 15.165 - j11.89$$

$$I_{aA} = I_1 = 24.34\angle -12.64° \ A$$

$$I_{bB} = I_2 - I_1 = 10.81\angle -142.6° \ A$$

$$I_{cC} = -I_2 = 19.27\angle 141.9° \ \text{A} \ .$$

Step 4 of 4

(B) $\overline{S_a} = (240\angle 0°)(24.34\angle 12.64°) = 5841.6\angle 12.64°$

$$\overline{S_b} = (240\angle -120°)(10.81\angle 42.6°) = 2834.4\angle 22.6°$$

$$\overline{S_c} = (240\angle 120°)(19.207\angle -141.9°) = 4624.8\angle -21.9°$$

$$\overline{S} = \overline{S_a} + \overline{S_b} + \overline{S_c}$$

$$= 12.386 + j0.58 \ KVA$$

$$= 12.4\angle 2.54° \ KVA \ .$$